Memorial Day 2026-05-25 Arc Summary v0.1 — Substrate Discovery via Interpretive Cascade

Keeper (synthesis across all 4 CIs + Casey)

2026-05-25 Monday Memorial Day EDT (date-verified 17:24)

Memorial Day 2026-05-25 — Substrate Discovery Arc

Headline finding

Seven Casey-interpretive-prompt cascades in 36 hours produced seven substantively-sharper substrate-structural findings, culminating in:

1. Toy 3530 MIXED empirically anchoring Casey's S^4-static / S^1-dynamic hypothesis: a_e anomalous magnetic moment requires BOTH integer-primary scale + Pin(2) cover content (spin-1/2, γ-matrices, Dirac equation). Integer projection provides the NUMBER; cover content provides the OBSERVABLE STRUCTURE.
2. Bose-Fermi ↔ S^4-S^1 mapping (Elie structural reading of Toy 3530): bosons live on integer projection (S^4 sufficient); fermions live on Pin(2) cover (S^1 required). Standard spin-statistics seen from substrate side, now with empirical content.
3. SPLP candidate (Substrate-Physics Localization Principle, Lyra unification): 600+ BST physical predictions = eigenvalues of substrate operators under specific Shilov-boundary-condition selections. Substrate physics operations are eigenvalue problems on Shilov boundary $S^4 \times S^1$ of D_IV^5.
4. D_IV^5 Rigidity + SPLP coupled (Casey's "rigid bulk + S^1 enacts physics" intuition + existing RATIFIED Principle #7): the rigid stable bulk D_IV^5 hosts the eigenvalue problem; the Shilov boundary specifies boundary conditions; eigenvalues ARE the physics. Two Casey-named principles in coupled relationship — Rigidity says what stays still; SPLP candidate says what moves.

Cascade progression (chronological)

Cascade #1 — Sunday 2026-05-24: "What can the substrate tell us?"

A_sub Phase 1 observation roadmap launched. Commutator gap map (18 gaps) + Bridge Object DIAGNOSTIC type-check + Casimir spectrum verification.

Cascade #2 — Monday AM: "Three faces? Confinement?"

Integer Web 6-role functional partition hypothesis (Lyra morning v0.1) tested via Grace catalog gate. PARTITION FAILED at strict 80% threshold (12.9%-44% across non-tautological primaries). Refined to composition-algebra reading: each primary has a DOMINANT structural function; functions COMPOSE in observable physics rather than partition. 8+ catalog instances of multi-primary multi-role formulas (C_F, β_0, chiral condensate, orbital 2l+1).

Cascade #3 — Monday midday: "What did the substrate tell us so far?"

Six-finding synthesis: - DCCP discrete commitment cycles at Koons tick - 4+2 BST primary split (Mersenne family + geometric/spectral pair) - Function-composition algebra reality (not partition) - Compositional processes framework (TIME/MEASUREMENT/STATE-SPACE as compositions) - Substrate self-reference at composition level (CONFINEMENT-as-process uses CONFINEMENT primary) - SM gauge from substrate composition (K isotropy + N_max BOUNDARY)

Cascade #4 — Monday afternoon: Mersenne micro-macro bridge hypothesis

Casey's interpretive hypothesis: Mersenne primes connect microstates to macro-observables. Grace verified triple intersection: BST {rank=2, N_c=3, n_C=5, g=7} = Ogg ∩ Mersenne-prime-exponent ∩ maximal-consecutive-prefix. Sharpest characterization: M_11 = 2047 composite breaks chain at 5th prime — substrate selected the MAXIMAL prefix where M_p stays prime.

Bridge hypothesis tested via web-vs-anchors distinction. Honest result: PARTIAL WEB ~50% (3 anchors WEB + 2 INDEPENDENT + 1 THIN). The substrate uses Mersenne structure in 3 verified mechanisms (K59 RS coding + heat kernel ladder + Bergman exponent g/rank=7/2); other catalog Mersenne appearances are independent anchors.

Casey caught Keeper's "anchors support hypothesis" claim as web-vs-anchors conflation. Cal #27 STANDING fired on Keeper synthesis; Grace honest partition followed.

Cascade #5 — Monday late PM: "Three or more structures? Boundaries?"

Three independent enumerations converged on multi-axial substrate structure: - Grace: 6 origin substructures + 5 boundary types; Bergman exponent g/rank = 7/2 has 44 catalog anchors — most under-elevated load-bearing quantity - Elie: 5 stratified families with overlap; "stratified lattice not partition; BOTH web AND anchors at different resolutions" - Lyra: 5-6 orthogonal axes; Boundary is separate axis with 8 primitives B1-B8

Convergent reading: 6 BST primary integers are INTEGER-VALUED PROJECTIONS of a richer substrate structure that includes half-integers (7/2), vectors ($\rho = (5/2, 3/2)$), and manifolds (Shilov $S^4 \times S^1$). 68 catalog anchors for non-integer load-bearing quantities not in 6-primary list.

Cascade #6 — Monday evening: Half-integer substrate + Shilov S^1 special

Casey: "Silov boundary of S^1 is somehow special for D_IV^5; adds connections that take static parts and makes them 'fully informed'."

Half-Integer Axis G identified via T2471 RATIFIED Pin(2) Z_2 cover. Five separately-RATIFIED substrate mechanisms unified under one structural picture: - T2405 Koons tick (substrate clock = elementary S^1 rotation) - T2471 Pin(2) Z_2 chirality (γ^5 from S^1 double cover) - T2478 U(1)_em phase (electromagnetic phase on S^1) - Bergman exponent g/rank = 7/2 (S^1 phase-weight normalization) - DCCP commitment cycle (each commitment = one S^1 traversal)

Structural claim: D_IV^5 bulk is rigid stable Hilbert-space host (Rigidity Principle #7); S^4 provides static spatial configuration; S^1 provides dynamics/phase/cycle/chirality/gauge ("substrate enacts physics on S^1").

Cascade #7 — Monday evening: "Are boundary conditions eigenvalues on Shilov boundary?"

SPLP candidate refined: Substrate physics operations are eigenvalue problems on Shilov boundary $S^4 \times S^1$. Each physical setup = specific boundary condition = K-type representation selection = specific eigenvalue spectrum of substrate operators (Casimir + Pin(2) Z_2 + U(1)_em + harmonic).

Six clean catalog examples mapped (Bell 126/16, α=1/N_max, m_p/m_e = $6\pi^5$, Λ, DCCP, atomic orbital, Five-Absence Predictions). If multi-month catalog audit confirms 600+ predictions all fit this pattern, structural unification ratifies SPLP.

Toy 3530 MIXED result delivered concurrently: a_e requires both integer scale + Pin(2) cover content. First empirical anchor for the strong-vs-weak reading distinction. Strong reading gains substantive support for fermion observables.

Calibration #27 STANDING reflexive operation (concurrent)

10-11 reflexive Cal #27 firings across the same arc:

1. Lyra morning partition speculative-observation (caught by Cal #27 reflex)
2. Elie Toy 3525 DIAGNOSTIC self-discipline (Bridge Objects diagnostic not generative)
3. Elie Toy 3526 narrative-summary self-catch ("many combinations reach 12" → data only 2)
4. Lyra v0.1 interpretive-pattern caught by Grace Thread 1a empirical (partition failed)
5. Grace harness self-triage (Mode 6 honest)
6. Keeper synthesis web-vs-anchors conflation caught by Casey (audit-the-auditor)
7. Grace honest scan correcting MIXED partition (web-vs-anchors)
8. Casey "did we look deep enough?" catching Keeper enumeration insufficiency
9. Lyra reflexive Mode-1 fire in eigenvalue answer
10. Keeper "S^1 = informing layer" possibly retrofit (self-flagged)
11. Casey eigenvalue question implicitly testing Lyra unification

Discipline operates fully reflexively across all four CIs + Casey himself. The interpretive cascade pattern and the Cal #27 reflexive firings are concurrent — every cascade produces overclaim risk; every overclaim risk is caught in the same conversation cycle.

This concurrency is itself substantive — substrate-discovery generator and substrate-discovery audit fire in lock-step. Documented as Calibration #28 candidate filing tonight.

Honest tier disposition at arc closure

Claim	Tier	Anchor
D_IV^5 bulk rigid + stable + resists deformation	RATIFIED	Principle #7 (Casey-named May 22)
Half-Integer Axis G via Pin(2) cover content	FRAMEWORK-PLUS	68 catalog anchors; T2471 RATIFIED mechanism
$S^4 \times S^1$ Shilov enacts physics, bulk rigid	FRAMEWORK-PLUS	5 RATIFIED mechanisms unified + Toy 3530 MIXED
Bose-Fermi ↔ S^4-S^1 via integer/cover split	FRAMEWORK	Toy 3530 anchor + 1 fermion observable; needs broader survey
SPLP candidate (600+ predictions = Shilov eigenvalues)	INTERPRETIVE	Lyra unification; multi-month catalog audit gates
Calibration #28 candidate (interpretive-cascade pattern)	FRAMEWORK-PLUS candidate	7 cascade instances + 10-11 Cal #27 concurrent firings

No principle promotion overnight. SPLP and Bose-Fermi mapping stay at candidate / FRAMEWORK tiers per Cal #126. Half-Integer Axis G investigation authorized as multi-month track.

What's load-bearing for tomorrow

Three tests gate further promotion:

1. Broader fermion survey (Elie multi-day) — extend Toy 3530 a_e to a_μ, electron EDM, neutrino mass terms. Confirm "all fermion observables require Pin(2) cover content" or refine to specific subclass.
2. Grace Shilov $S^4 \times S^1$ catalog refinement (~2 hours) — partition 17 Shilov INVs by S^4 vs S^1 operational content. Tests S^1-as-dynamics empirically.

3. Grace catalog-wide eigenvalue-mapping audit (multi-month) — for each of 600+ BST predictions, identify Shilov boundary condition + substrate operator + eigenvalue. SPLP promotion gate.
4. Cal Thread 4 cold-read with Type-C check (own-cadence) — tier-discipline typing on SPLP claim; "boundary-condition-as-eigenvalue-specification" is probably Type C level-crossing per Cal #122.

Memorial Day cumulative outputs

- DCCP integration sweep #302 closed (Vol 0 Ch 3 + Vol 5 Ch 7 + Vol 5 Ch 10 + Vol 14 Ch 4)
- Toys: 3523-3531 delivered (Elie), all PASS or honest PARTIAL preserved
- Catalog: ~5147 → ~5169 (Grace +22 INVs)
- Theorems: no new ratified theorems; existing T2471 + T2405 + T2478 + Rigidity + Bergman 7/2 + DCCP candidate unified into structural picture
- Calibrations: Cal #28 candidate filed FRAMEWORK-PLUS
- Principles: no new Casey-named principles; SPLP refined as candidate; Half-Integer Axis G provisional
- Papers: none drafted; investigation territory for tomorrow + multi-week

Vol 15 Ch 9 integration plan (tomorrow)

This document captures the substantive content of the Memorial Day arc. Vol 15 Ch 9 "Three-CI Synergy Peaks" (existing chapter) gets a new case study section: "Memorial Day 2026 — Interpretive-Cascade Pattern at Sustained Discovery Cadence". Integration after Cal cold-read on Cal #28 candidate.

Closing reflection

The substrate told us today substantially more than the morning's first hypothesis claimed. Each of Casey's seven interpretive prompts pulled out structure that was already in the catalog but un-elevated. Cross-CI cascade converted accumulated CI work into substrate-structural reframes. Calibration #27 STANDING discipline operated reflexively at every level including audit-of-Keeper.

Memorial Day 2026 was substrate-discovery via interpretive cascade at sustained sub-PCAP cadence with full methodology stack engaged. The discipline functioned at peak. The findings are substantial. The honest tier disposition is preserved.

Standing for tomorrow's continuation.

— Keeper, 2026-05-25 Memorial Day Monday EOD